

SoC DESIGN – WEEK 7

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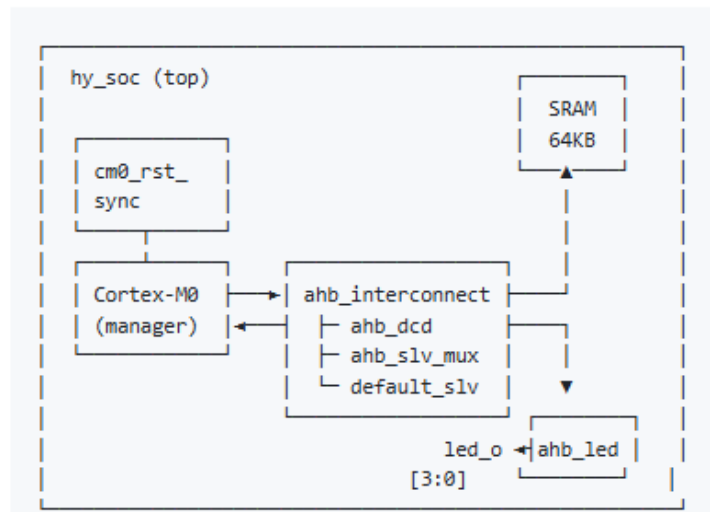
TA

Professor: Ji-Hoon Kim

Lab2

- Cortex-M0 + LED Controller
 - Builds on Lab 1 by adding an LED controller peripheral, implementing SysTick interrupt-driven LED blinking. The AHB interconnect is expanded to handle two subordinates.
 - Peripheral Addition Pattern: Add address region to DCD → expand MUX → instantiate in top
 - SysTick Interrupt: CM0 built-in timer for periodic LED toggling
 - Simulation Mode: SIMULATION=1 macro shortens the blink period for faster simulation

Block Diagram



Changes from Lab 1

Type	Description
New	<code>ahb_led</code> — LED controller (4-bit output)
Modified	<code>hy_soc.v</code> — Added <code>led_o[3:0]</code> port and LED instance
Modified	<code>ahb_dcd.v</code> — Added LED address region (2-to-1 decoding)
Modified	<code>ahb_slv_mux.v</code> — Expanded to 3-way mux (SRAM + LED + Default)

Directory Structure

- Cortex-M0 + LED Controller

Directory Structure

```
lab2_m0_led/
├── rtl/
│   ├── hy_soc.v          ← Top-level (led_o port added)
│   ├── ahb_interconnect.v ← Interconnect (2 subordinates)
│   ├── ahb_dcd.v         ← Address decoder (SRAM + LED)
│   └── ahb_slv_mux.v      ← 3-way response mux
├── tb/
│   └── tb_hy_soc.v        ← Testbench (LED output monitoring)
├── sw/
│   ├── test.c            ← SysTick LED blink test
│   ├── gcc/              ← GCC build
│   └── arm/              ← ARM Compiler build
└── Makefile
```

Memory Map

Address Range	Device	Notes
0x0000_0000 – 0x0000_FFFF	SRAM (64KB)	haddr[31:16] == 16'h0000
0x5000_0000 – 0x50FF_FFFF	LED Controller	haddr[31:24] == 8'h50
All other	Default Slave	ERROR response

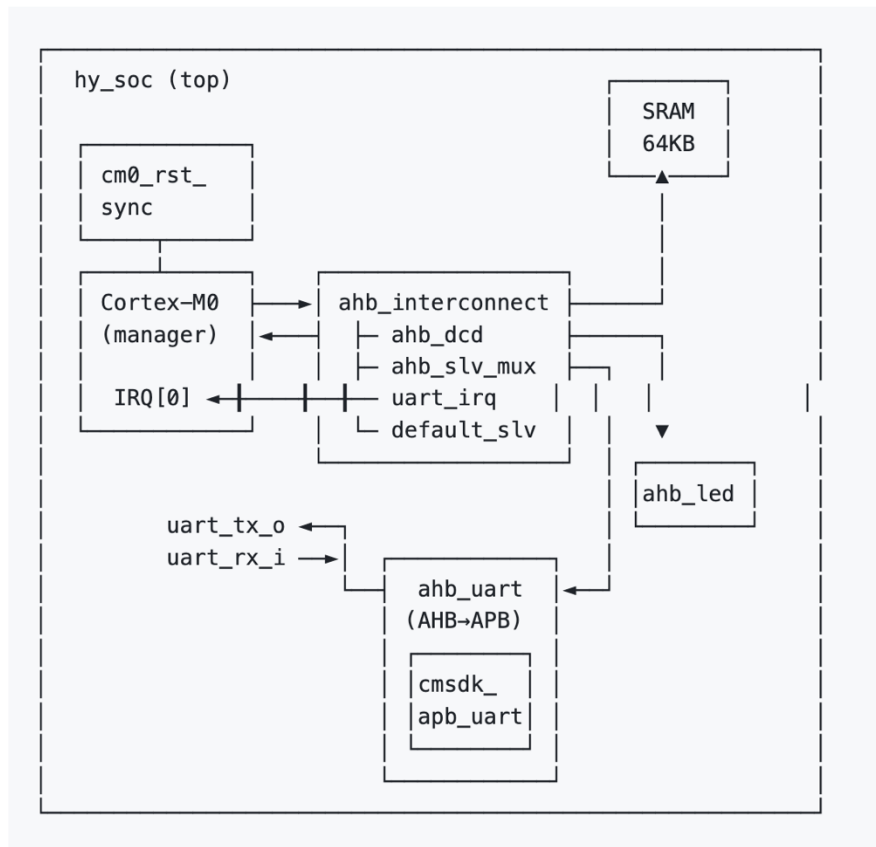
LED Registers

Offset	Name	Description
0x00	DATA	LED output [3:0] (R/W)

Lab3

- Cortex-M0 + UART
 - Builds on Lab 2 by adding a **CMSDK APB UART** peripheral, enabling serial communication. Introduces the AHB-to-APB bridge pattern and interrupt routing.

Block Diagram



Changes from Lab 2

Type	Description
New	ahb_uart — AHB-Lite to APB adapter wrapper (1 wait-state FSM)
New	cmsdk_apb_uart — CMSDK-compatible UART (Verilator-clean rewrite)
Modified	hy_soc.v — Added UART TX/RX ports and interrupt (IRQ[0]) routing
Modified	ahb_dcd.v — Added UART address region (3-to-1 decoding)
Modified	ahb_slv_mux.v — Expanded to 4-way mux

Directory Structure

- Cortex-M0 + UART

Directory Structure

```
lab3_m0_uart/
├── rtl/
│   ├── hy_soc.v          ← Top-level (UART TX/RX/IRQ added)
│   ├── ahb_interconnect.v ← Interconnect (3 subordinates)
│   ├── ahb_dcd.v         ← Address decoder (SRAM + LED + UART)
│   └── ahb_slv_mux.v      ← 4-way response mux
├── tb/
│   └── tb_hy_soc.v        ← Testbench (UART loopback + uart_mon)
├── sw/
│   ├── test.c            ← SRAM + LED + UART integration test
│   ├── gcc/              ← GCC build
│   └── arm/              ← ARM Compiler build
├── fpga/
│   ├── nexys_a7/         ← Nexys A7-100T (USB-UART on-board)
│   │   ├── top_fpga.v    ← Board-specific top wrapper
│   │   ├── nexys_a7.xdc  ← Pin constraints
│   │   └── create_project.tcl
│   └── zybo_z7/          ← Zybo Z7-20 (UART via Pmod JE)
│       ├── top_fpga.v
│       ├── zybo_z7_20.xdc
│       └── create_project.tcl
├── host/
│   └── hy_term.py         ← Simple serial terminal for FPGA UART
└── Makefile
```

Memory Map

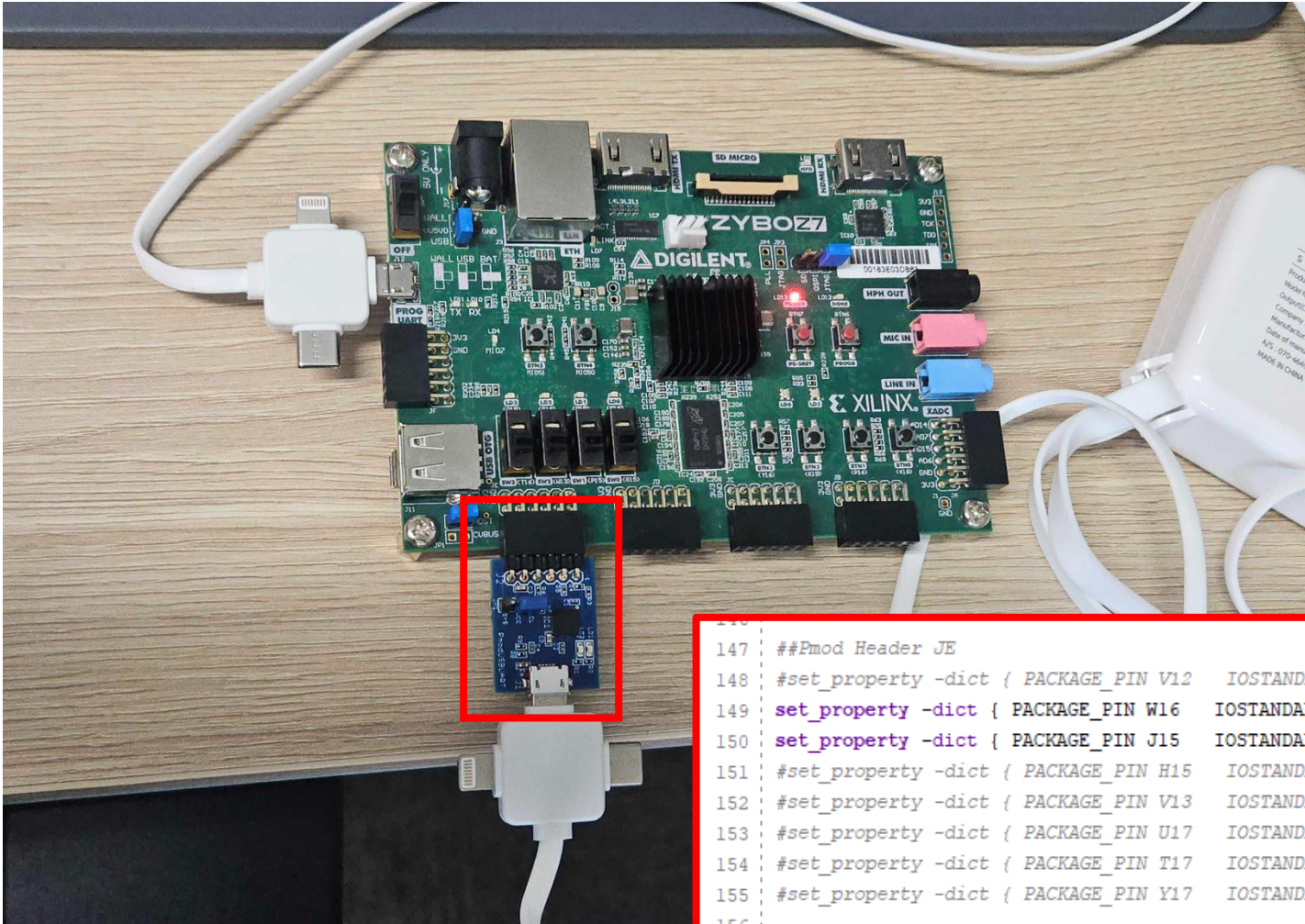
Address Range	Device	Notes
0x0000_0000 – 0x0000_FFFF	SRAM (64KB)	haddr[31:16] == 16'h0000
0x5000_0000 – 0x50FF_FFFF	LED Controller	haddr[31:24] == 8'h50
0x5100_0000 – 0x51FF_FFFF	UART	haddr[31:24] == 8'h51
All other	Default Slave	ERROR response

UART Registers (Base: 0x5100_0000)

Offset	Name	Description
0x00	DATA	W: TX data [7:0], R: RX data [7:0]
0x04	STATE	[3:0] = {rx_ovr, tx_ovr, rx_full, tx_full}
0x08	CTRL	[6]=HSTM [5:4]=ovr_int_en [3:2]=int_en [1:0]=en
0x0C	INTSTATUS	R: interrupt status / W: interrupt clear
0x10	BAUDDIV	[19:0] baud divider (minimum 16)

Directory Structure

- FPGA Board + Pmod_UART

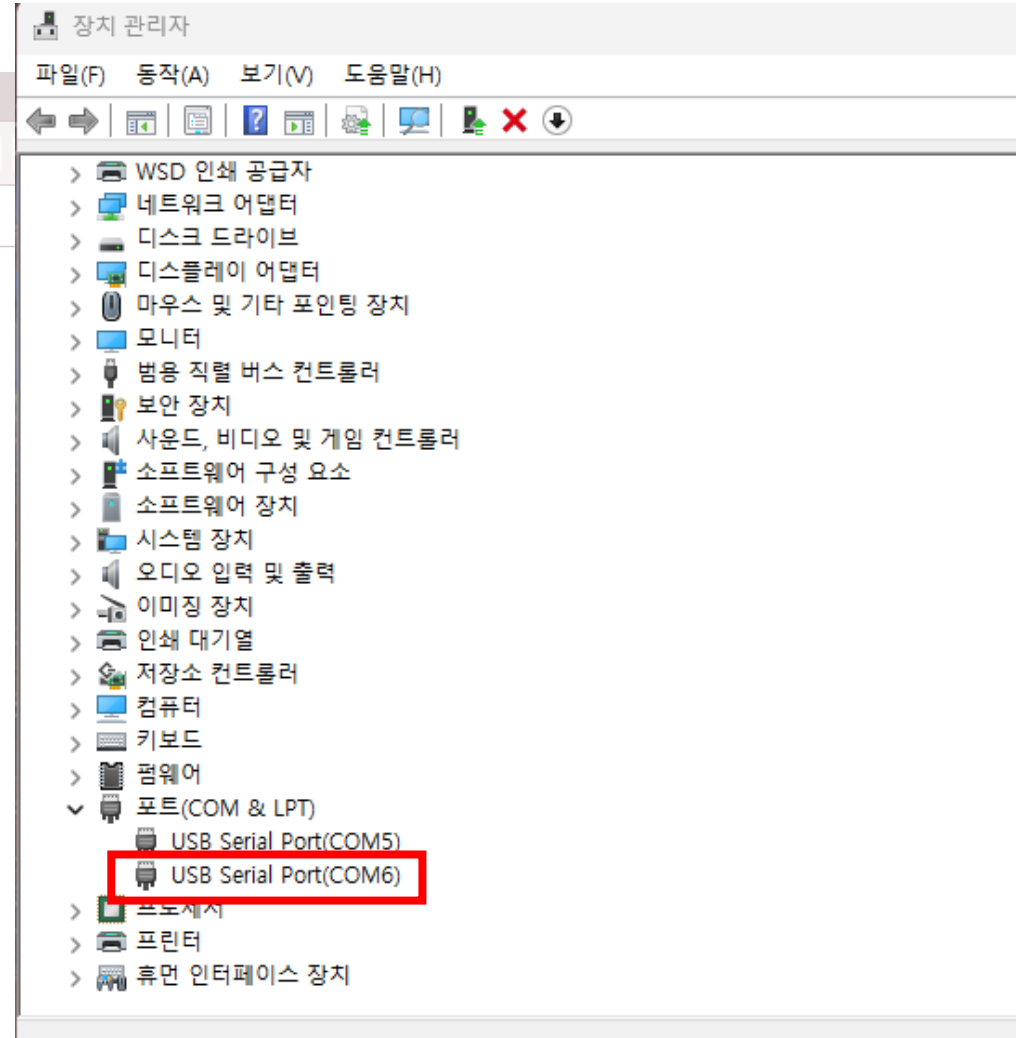
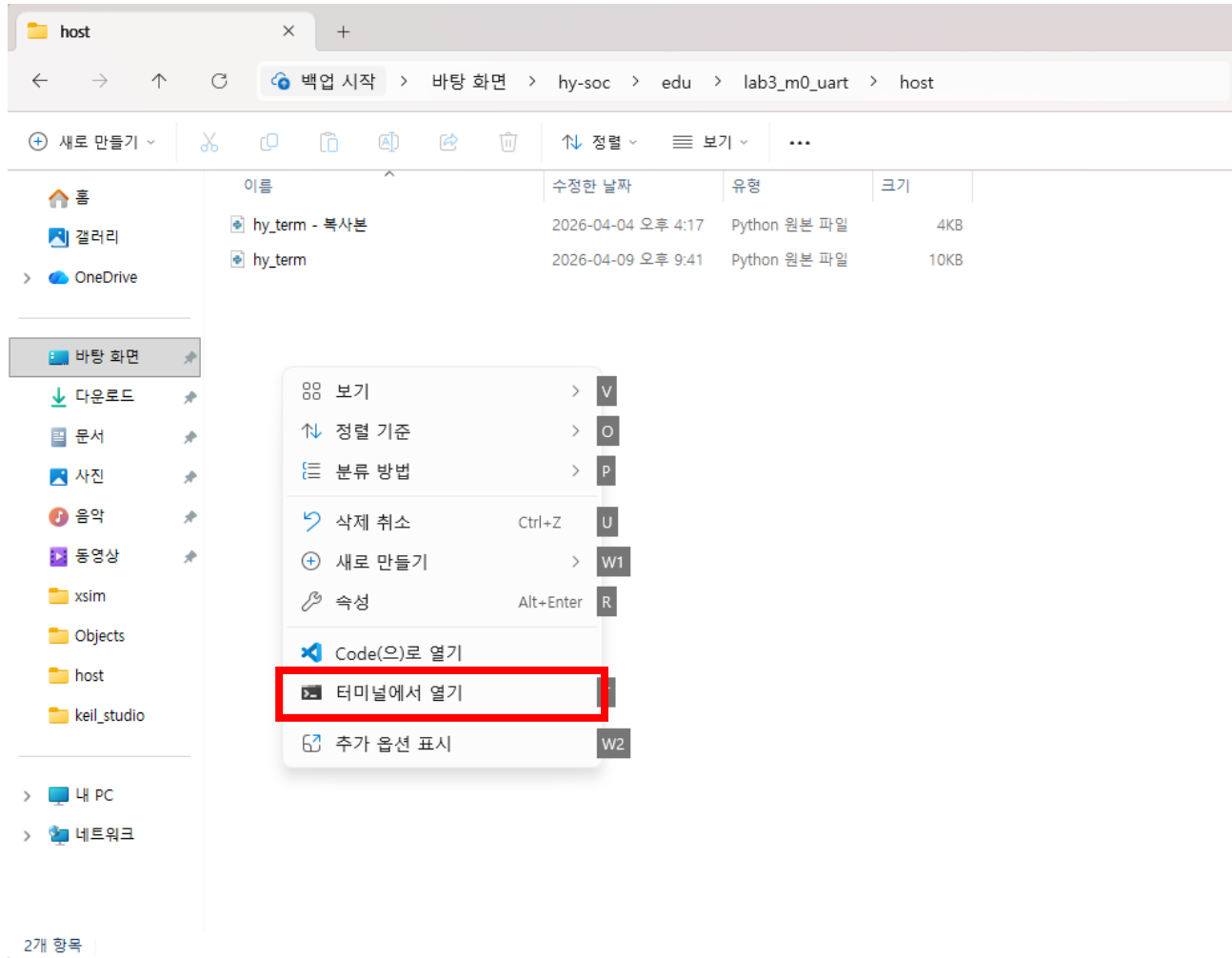


.xdc file

```
147 ##Pmod Header JE
148 #set_property -dict { PACKAGE_PIN V12 IOSTANDARD LVCMOS33 } [get_ports { je[0] }]; #IO_L4P_T0_34 Sch=je[1]
149 set_property -dict { PACKAGE_PIN W16 IOSTANDARD LVCMOS33 } [get_ports { pmod_uart_tx }]; #IO_L18N_T2_34 Sch=je[2]
150 set_property -dict { PACKAGE_PIN J15 IOSTANDARD LVCMOS33 } [get_ports { pmod_uart_rx }]; #IO_25_35 Sch=je[3]
151 #set_property -dict { PACKAGE_PIN H15 IOSTANDARD LVCMOS33 } [get_ports { je[3] }]; #IO_L19P_T3_35 Sch=je[4]
152 #set_property -dict { PACKAGE_PIN V13 IOSTANDARD LVCMOS33 } [get_ports { je[4] }]; #IO_L3N_T0_DQS_34 Sch=je[7]
153 #set_property -dict { PACKAGE_PIN U17 IOSTANDARD LVCMOS33 } [get_ports { je[5] }]; #IO_L9N_T1_DQS_34 Sch=je[8]
154 #set_property -dict { PACKAGE_PIN T17 IOSTANDARD LVCMOS33 } [get_ports { je[6] }]; #IO_L20P_T3_34 Sch=je[9]
155 #set_property -dict { PACKAGE_PIN Y17 IOSTANDARD LVCMOS33 } [get_ports { je[7] }]; #IO_L7N_T1_34 Sch=je[10]
156
```

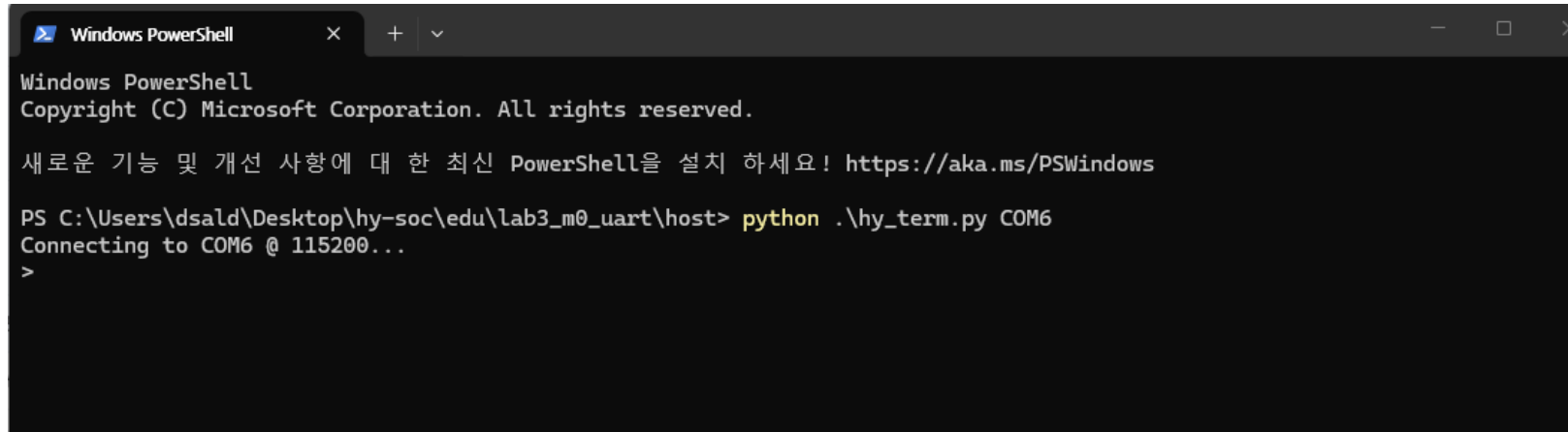
Directory Structure

• UART



Directory Structure

- Terminal > python .\hy_term.py **COM6**

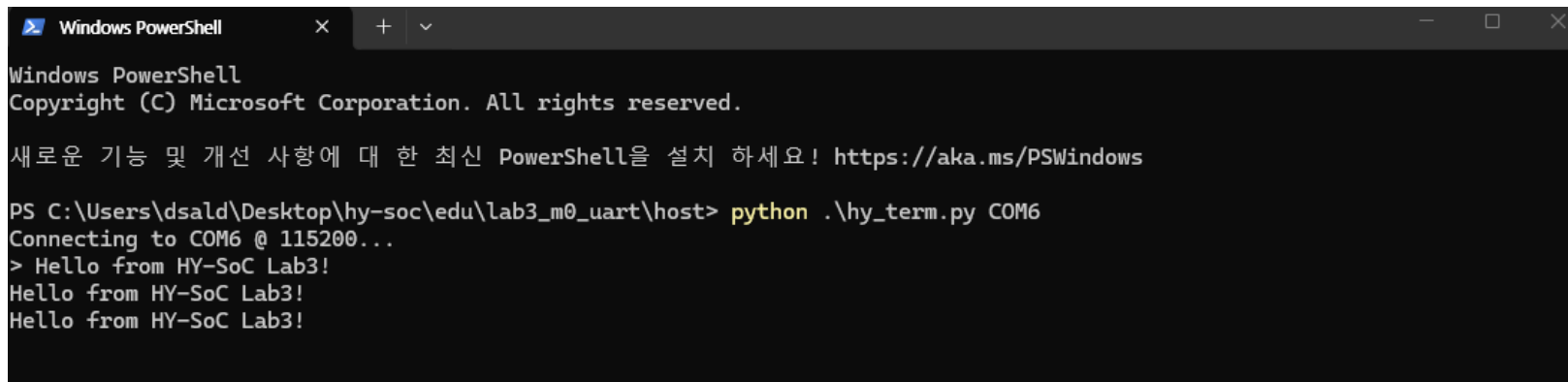


```
Windows PowerShell
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새로운 기능 및 개선 사항에 대 한 최신 PowerShell을 설치 하세요! https://aka.ms/PSWindows

PS C:\Users\dsald\Desktop\hy-soc\edu\lab3_m0_uart\host> python .\hy_term.py COM6
Connecting to COM6 @ 115200...
>
```

- Press **Btn0**



```
Windows PowerShell
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새로운 기능 및 개선 사항에 대 한 최신 PowerShell을 설치 하세요! https://aka.ms/PSWindows

PS C:\Users\dsald\Desktop\hy-soc\edu\lab3_m0_uart\host> python .\hy_term.py COM6
Connecting to COM6 @ 115200...
> Hello from HY-SoC Lab3!
Hello from HY-SoC Lab3!
Hello from HY-SoC Lab3!
```

To-Do

- **Lab2**

- FPGA Board Demo
 - Provided Code : reset 이후 (4'b0000), 4'b1111 -> 4'b0000 -> 4'b1111 -> 4'b0000
 - To-Do **Keil** : reset 이후, Blink 횟수 10회

- **Lab3**

- FPGA Board Demo
 - Provided Code : reset 이후, "Hello from HY-SoC Lab3!" 출력
 - To-Do **Keil** : reset 이후, "(본인학번)_(본인이름)" 출력

- **"To-Do"**

- Simulation Waveform 캡처 및 설명
- Code에서 수정한 부분 캡처 및 설명
- FPGA Board 동작에 대한 설명
 - (Lab3의 경우 terminal 결과 캡처)

THANK YOU

Any questions or comments:  Issu07@hanyang.ac.kr